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Maastricht University

Faculty of Sciences ***School for Information Technology***

Master of Statistics and Data Science

Master's thesis

Prediction of catheter's tip position based on Xray images of renal-disease patients subjected to dialysis.

Apu Chandra Bhowmick

Thesis presented in fulfillment of the requirements for the degree of Master of Statistics and Data Science,
specialization Data Science

SUPERVISOR :

Prof. dr. Dirk VALKENBORG

SUPERVISOR :

Tomasz HRYSZKO

Transnational University Limburg is a unique collaboration of two universities in two countries: the University of Hasselt and Maastricht University.



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Abstract

Central venous catheters (CVC) are life supporting devices that require correct positioning of their tips within the right atrium to be clinically safe. Malpositioned tips can result in serious complications, including cardiac arrhythmia and inadequate hemodialysis due to insufficient blood flow. Radiographic verification remains the standard of care, but manual review of radiologists is time-consuming and subject to variability.

This thesis proposes an automated two-step classification system that extracts whether a CVC tip is correctly positioned or malpositioned from chest X-ray segmentation masks. Extending the EfficientNet-B7 [13] full-catheter-mask approach of Forgwa [5] contributes two methodological points: (1) a new tip detection algorithm that finds the free end of the catheter skeleton via a thickness-based rule, and (2) a tip-dot EfficientNet-B7 model that encodes the tip as a Gaussian dot overlaid on the atrium mask instead of giving the whole catheter shape.

The 200 images with sure tip detection were passed to the tip-dot model, and the 20 images with ambiguous tip detection were passed to Forgwa’s full-mask model in a two-step classifier. The proposed system was evaluated on 220 chest X-ray images through five-fold stratified cross-validation, with an out-of-fold AUC of **0.965**, accuracy of **0.914**, sensitivity of **0.931** and specificity of **0.899** - improvements of +0.035 and +0.055 in AUC and accuracy over the full-mask baseline. Classical machine learning classifiers based on two hand-crafted tip-point features (normalised tip depth in the atrium, catheter radius at the tip) obtain AUC up to 0.900, confirming that tip location alone contains strong discriminative information.

Keywords: central venous catheter, chest X-ray, EfficientNet-B7, transfer learning, tip detection, binary classification, two-step classifier

Contents

Abstract	1
1 Introduction	4
1.1 Clinical Background	4
1.2 Prior Work and Motivation	4
1.3 Research Questions	5
1.4 Thesis Structure	5
2 Dataset	6
2.1 Image Collection	6
2.2 Class Distribution	6
2.3 Segmentation Masks	7
3 Tip Detection and Feature Engineering	8
3.1 Tip Detection Algorithm	8
3.1.1 Binarization	8
3.1.2 Skeletonization	8
3.1.3 Loop Detection and Recovery	9
3.1.4 Tip Selection: the Thickness Rule	10
3.2 Tip Detection Status	11
3.3 Candidate Tip-Point Features	12
3.4 Correlation Analysis and Feature Selection	13
4 Classical Machine Learning Methods	17
4.1 Experimental Design	17
4.2 Logistic Regression	17
4.3 Random Forest	17
4.4 XGBoost	18
4.5 Feature Importance	18
4.6 Results	19
5 EfficientNet-B7 with Tip-Dot Input	21
5.1 Motivation and Design	21
5.2 Model Architecture	22
5.3 Training Configuration	22
5.4 Results	23
5.5 Error Analysis by Tip Detection Status	24
6 EfficientNet-B7 with Full Catheter Mask (Baseline)	25

6.1	Architecture and Input	25
6.2	Training Configuration	25
6.3	Results	26
7	Two-Step Classifier	27
7.1	Design Rationale	27
7.2	Ensemble Prediction	27
7.3	Results	27
8	Full Comparison and Improvement Summary	29
8.1	Improvement over the Baseline	29
9	Discussion	31
9.1	Interpretation of Results	31
9.2	Limitations	32
9.2.1	Small single-centre dataset	32
9.2.2	Dependency on segmentation quality	32
9.2.3	Potential inter-rater variability	32
9.2.4	Frontal images only	32
9.2.5	Limited exploration of alternative approaches	33
9.2.6	No end-to-end training	33
9.3	Future Work	33
10	Ethical Thinking, Societal Relevance, and Stakeholder Awareness	34
10.1	Ethical Aspects	34
10.1.1	Data and patient privacy	34
10.1.2	Informed consent	34
10.1.3	Honest reporting	34
10.1.4	Algorithmic fairness and bias	34
10.2	Societal Relevance	35
10.3	Stakeholder Awareness	35
11	Conclusion	36
	Generative AI Acknowledgement	39
A	Software and Code	40

Introduction

Clinical Background

Central venous catheters (CVC) are life supporting devices that need to be correctly positioned at the tips within the right atrium to be clinically safe. Malpositioned tips can result in serious complications, including cardiac arrhythmia and inadequate hemodialysis due to insufficient blood flow [10]. Radiographic verification is the standard of care but manual review of radiologists is time-consuming and prone to variability.

This thesis develops an automated two-step classification system that identifies, from chest X-ray segmentation masks, whether a CVC tip is correctly placed or malpositioned. Building upon the EfficientNet-B7 [13] full-catheter-mask approach of Forgwa [5] makes two methodological contributions: (1) a novel tip detection algorithm that locates the free end of the catheter skeleton using a thickness-based rule, and (2) a tip-dot EfficientNet-B7 model that encodes the tip as a Gaussian dot superimposed on the atrium mask rather than providing the whole catheter shape.

Prior Work and Motivation

Forgwa [5] developed a UNet++ segmentation pipeline that produces binary masks of the catheter tube and right atrium from chest X-ray images. Feeding these masks to an EfficientNet-B7 deep learning model, Forgwa achieved an out-of-fold AUC of 0.927 on a dataset of 220 images from the Białystok Medical University (Poland). While this result is promising, several limitations motivated the present work:

1. The model received the *full catheter mask* - a complex, variable shape - even though the clinically relevant information is primarily the *location of the tip* relative to the atrium.
2. No attempt was made to localise the free endpoint of the catheter from the segmentation mask. The spatial relationship between tip and atrium was not explicitly encoded.
3. Classical machine learning approaches using interpretable tip-point features were not explored, leaving a gap in the explanation of which anatomical signals drive classification.

This thesis addresses all three limitations through a combined feature engineering and deep learning approach.

Research Questions

This thesis investigates the following research questions:

1. Can a skeleton-based tip detection algorithm reliably locate the free endpoint of the catheter from the predicted segmentation mask, and how often does this fail?
2. Do tip-point features (particularly the normalised depth of the tip within the right atrium) provide discriminative power for classical machine learning classifiers?
3. Does an EfficientNet-B7 model trained on a Gaussian dot representation of the tip location outperform classical ML and approach the performance of the full-mask model?
4. Does a two-step routing strategy - using the tip-dot model when the tip is reliably detected and falling back to the full-mask model for ambiguous cases - improve overall classification performance?

Thesis Structure

Section 2 describes the dataset. Section 3 explains the tip detection algorithm and feature engineering pipeline. Section 4 presents the classical machine learning methods. Section 5 describes the tip-dot EfficientNet-B7 model. Section 6 presents Forgwa’s full-mask baseline. Section 7 introduces the two-step classifier. Section 8 compares all methods. Section 9 discusses results, limitations, and future directions. Section 10 addresses ethical considerations.

Dataset

Image Collection

The dataset consists of **220 posterior-anterior chest X-ray images** of patients on dialysis who were treated at the Białystok Medical University (Poland). All photos were obtained as part of standard clinical care following CVC placement. Radiologists at the hospital assessed each case as correctly positioned or malpositioned based on the position of the catheter tip relative to conventional anatomical landmarks.

Class Distribution

Table 1: Class distribution of the 220-image dataset.

Class	n	Proportion
Malpositioned (label = 0)	119	54.1 %
Correctly placed (label = 1)	101	45.9 %
Total	220	100 %

The mild class imbalance (approximately 54/46) was accounted for in model training where relevant.

Figure 1 shows one example from each class. The difference between a correctly placed and a malpositioned catheter comes down to where the tip ends up relative to the right atrium.

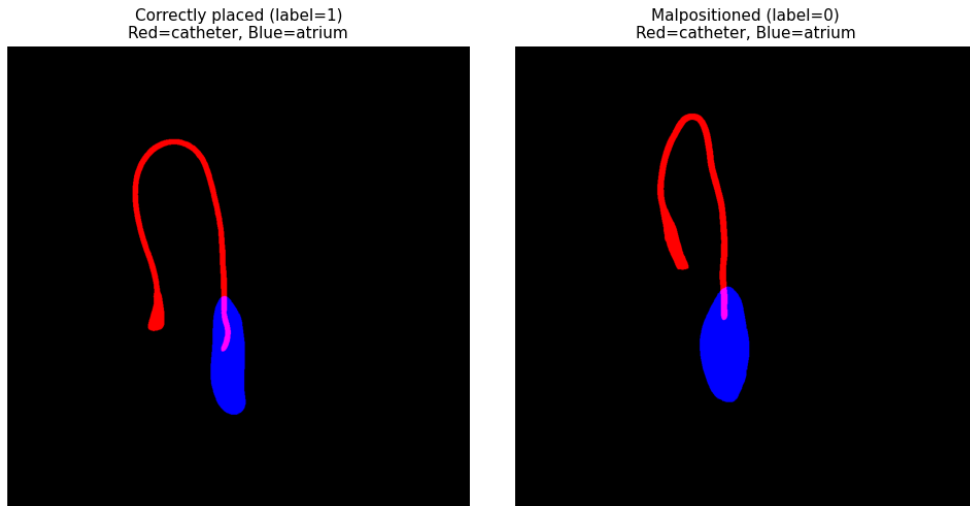


Figure 1: Example catheter masks from the dataset. Left: correctly placed catheter (label = 1). Right: malpositioned catheter (label = 0). The atrium mask overlay (shown in the segmentation output, Figure 2) provides the anatomical reference for tip position assessment.

Segmentation Masks

Binary segmentation masks of the catheter tube and the right atrium were created for each picture using the UNet++ [14] pipeline of Forgwa [5]. These masks, rather than the original radiograph pixel values, form the input to all models in this thesis. This design decision distinguishes the segmentation and classification problems, which allows the evaluation of classification algorithms independently of the quality of the raw images.

Each mask is saved as a 960×960 greyscale PNG picture. A value of a pixel larger than zero indicates that the matching structure is present.

Figure 2 displays the two input masks and their overlay. All models in this thesis work from these masks rather than the raw X-ray.

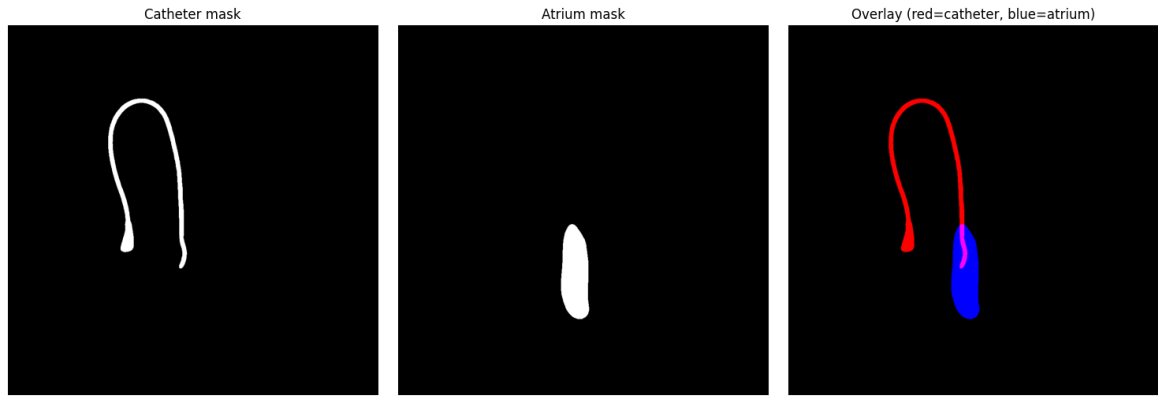


Figure 2: UNet++ [14] segmentation output. Left: catheter mask. Centre: atrium mask. Right: overlay with catheter in red and atrium in blue.

Tip Detection and Feature Engineering

This section describes the two-stage pipeline that converts the raw catheter segmentation mask into the tip coordinate (`tip_y`, `tip_x`) and subsequently derives numerical features for classical machine learning.

Tip Detection Algorithm

Binarization

The catheter mask from UNet++ [14] is a continuous greyscale image. It is converted to a clean binary mask via:

1. **Gaussian blur** (5×5 kernel) to reduce pixel-level noise before thresholding.
2. **Otsu thresholding** [11], which automatically finds the optimal intensity threshold to separate foreground (catheter pixels) from background.
3. **Morphological opening** with a 2-pixel disk, which removes small boundary protrusions that would create false junctions during skeletonization.
4. **Morphological closing** with a 4-pixel disk, which fills small holes within the catheter body.
5. **Removal of small objects** (fewer than 50 pixels) to eliminate noise blobs.
6. **Largest connected component selection** to retain only the main catheter body if UNet++ produced multiple disconnected regions.

Skeletonization

The binary catheter mask is reduced to a single-pixel-wide centreline using the *medial axis transform* [8]. Simultaneously, the distance transform $d(p)$ is computed, which gives for every skeleton pixel p the local radius of the catheter at that point - i.e., the distance from p to the nearest background pixel.

Skeleton pixels are classified by their 8-connectivity degree:

- **Degree 1** (one neighbour): endpoint. A clean catheter has exactly two endpoints.
- **Degree 2** (two neighbours): midpoint along the centreline.
- **Degree ≥ 3** (three or more neighbours): junction, indicating the catheter crosses itself.

Loop Detection and Recovery

Some catheters form a loop where the tube curves back and crosses over itself, creating graph cycles in the skeleton (illustrated in Figure 3). Cycles are counted using Euler’s formula:

$$\text{cycles} = E - V + C \quad (1)$$

where E is the number of edges, V the number of vertices, and C the number of connected components in the skeleton graph. A cycle count greater than zero indicates the skeleton contains a closed loop.

To distinguish genuine catheter loops from small artefacts at the catheter cuff, a ratio test is applied: a loop is considered genuine only if the number of skeleton pixels participating in the cycle satisfies both:

- cycle pixels / total skeleton pixels ≥ 0.10 , and
- cycle pixels ≥ 80 .

When a genuine loop is detected, three recovery strategies are attempted in order:

1. **Junction cut:** erase a small disk around the junction point closest to the thickest endpoint.
2. **2-core boundary cut:** erase around the thickest point on the boundary of the loop’s 2-core.
3. **Curvature cut:** locate the point of maximum curvature along the longest skeleton path and cut there.

If all three strategies fail, or if the resulting skeleton still does not yield exactly two endpoints, the image is assigned status **ambiguous_tip** and is not processed further by the tip selection step.

Figure 3 illustrates the full process on a real case. Panel (a) shows the binary catheter mask. Panel (b) highlights the cycle pixels in red, confirming the skeleton forms a closed loop. Panel (c) shows the skeleton after the junction cut, with two clean endpoints recovered. Panel (d) marks the final tip selected by the thickness rule after recovery.

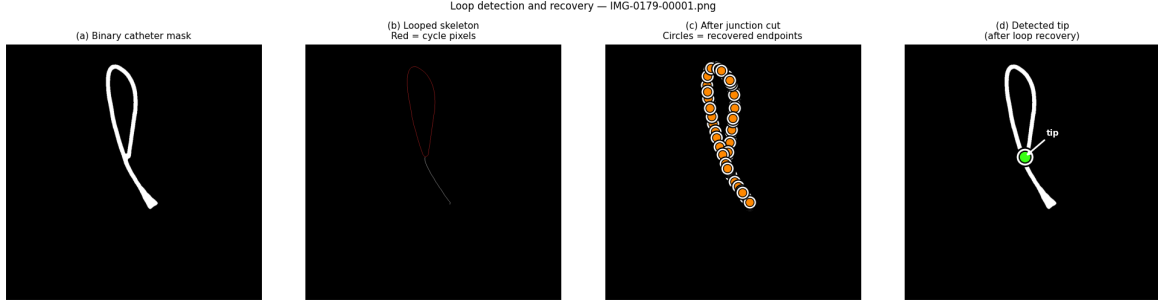


Figure 3: Loop detection and recovery on a `success_overlap_mode` case. (a) Binary catheter mask. (b) Skeleton with cycle pixels highlighted in red. (c) Skeleton after junction cut; circles mark the two recovered endpoints. (d) Final tip selected by the thickness rule after recovery.

Tip Selection: the Thickness Rule

For images with exactly two endpoints, the clinical tip is identified by the following rule:

The tip is the thinner endpoint.

Clinically, the free end of the catheter inside the heart is thin and flexible, while the connector end (where the catheter exits the body) is surrounded by a thick hub. The distance transform value $d(\text{endpoint})$ quantifies the catheter radius at each endpoint. The endpoint with the smaller d value is selected as the tip.

When the two endpoints have similar thickness (ratio > 0.90 , gap < 5 pixels), two tiebreakers are applied in order:

1. **Atrium proximity:** the endpoint closer to or more deeply inside the atrium mask is preferred.
2. **Vertical position:** the lower endpoint (larger y value in image coordinates) is preferred.

Figure 4 shows the full pipeline on one example: the binary mask, the skeleton with its two endpoints, and the final tip selected by the thickness rule.

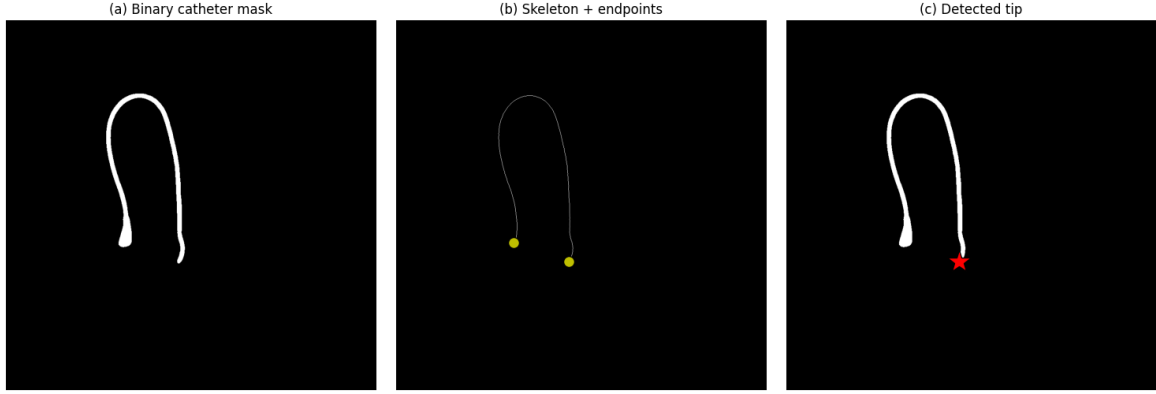


Figure 4: Tip detection pipeline. (a) Binary catheter mask. (b) Skeleton with the two endpoints marked in yellow. (c) The thinner endpoint selected as the tip, marked with a red star.

Tip Detection Status

After the algorithm runs, each of the 220 images receives one of three status labels:

Table 2: Tip detection status distribution ($n = 220$).

Status	Description	n	%
success	Clean catheter; tip found via thickness rule	197	89.5
success_overlap_mode	Loop recovered; tip found after cutting	3	1.4
ambiguous_tip	Recovery failed; tip unreliable	20	9.1

The status label is used *only* as a routing signal for the two-step classifier (Section 7); it is not used as a model feature.

Figure 5 shows one example of each status. The ambiguous case is the one the tip-dot model cannot handle reliably, which is the reason for the two-step design in Section 7.

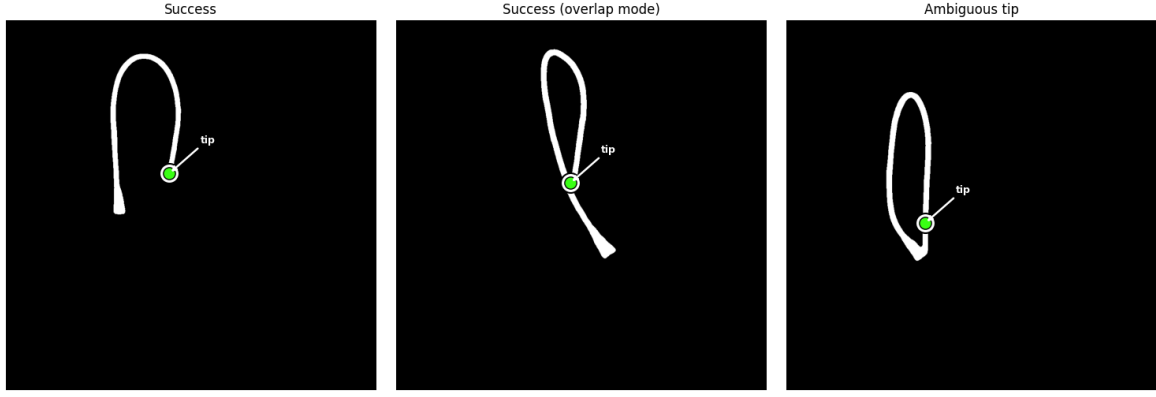


Figure 5: Examples of the three tip detection statuses. Left: a clean success case with the detected tip marked by a green circle. Centre: a looped catheter recovered in overlap mode with the tip marked. Right: an ambiguous case where the tip could not be reliably determined and no marker is shown.

Candidate Tip-Point Features

All features are derived from the single detected tip pixel (`tip_y`, `tip_x`) rather than from a segment of the catheter tube. Seven candidate features were computed. The *optimal depth zone* referred to below is defined as the vertical band between the upper-third line and the midpoint of the right atrium bounding box - the clinically accepted target range for a correctly positioned CVC tip.

distance_from_optimal_depth Signed continuous distance from the tip's vertical position to the optimal depth zone. Negative = tip too shallow (above the zone); positive = tip too deep (below the zone).

tip_depth_in_atrium_norm Arc-length depth of the tip along the atrium centre-line, normalised by atrium height (0 = top of atrium, 1 = bottom).

tip_y_norm_to_atrium Normalised vertical position of the tip within the atrium bounding box, computed as:

$$\frac{y_{\text{tip}} - y_{\text{atrium,top}}}{h_{\text{atrium}}}$$

where h_{atrium} is the atrium bounding-box height.

tip_x_norm_to_atrium Normalised horizontal position of the tip within the atrium bounding box, computed analogously using the atrium bounding-box width.

tip_to_atrium_dist_norm Euclidean distance from the tip to the nearest atrium

pixel, normalised by atrium height (equals 0 when the tip lies inside the atrium).

mean_tip_radius

Catheter thickness at the tip: the distance-transform value $d(\text{tip})$ in pixels, i.e. the distance from the tip skeleton pixel to the nearest background pixel, which equals the local half-width of the catheter.

tip_dy

Vertical component of the unit direction vector along the skeleton at the tip pixel, computed from the two nearest skeleton neighbours. A value near +1 indicates the catheter arrives at the tip moving straight downward; near 0 indicates a horizontal approach.

Correlation Analysis and Feature Selection

A Pearson correlation analysis was conducted to remove redundant features before model training, correlated features hurt logistic regression interpretability. Any pair with $|r| > 0.75$ was flagged. Where several features were mutually correlated, they were treated as a single group, and only the feature most strongly correlated with the binary classification label (correct placement vs. malposition) was retained.

Table 3: Highly correlated feature pairs ($|r| > 0.75$). In each correlated group, the feature with the strongest point-biserial correlation with the classification label was retained.

Feature A	Feature B	r	Retained
dist_from_optimal_depth	tip_y_norm_to_atrium	+0.982	tip_depth_in_atrium
dist_from_optimal_depth	tip_depth_in_atrium_norm	+0.772	tip_depth_in_atrium
dist_from_optimal_depth	tip_to_atrium_dist_norm	-0.767	tip_depth_in_atrium
tip_y_norm_to_atrium	tip_depth_in_atrium_norm	+0.838	tip_depth_in_atrium

*Feature names abbreviated: $\text{dist_from_optimal_depth} = \text{distance_from_optimal_depth}$;
 $\text{tip_depth_in_atrium} = \text{tip_depth_in_atrium_norm}$.*

These features form a single correlated cluster, all measuring the vertical depth of the tip within the atrium. Rather than making pairwise drops, the feature with the strongest point-biserial correlation with the classification label, **tip_depth_in_atrium_norm** ($r = +0.495$), was retained, and the other depth-related features were dropped.

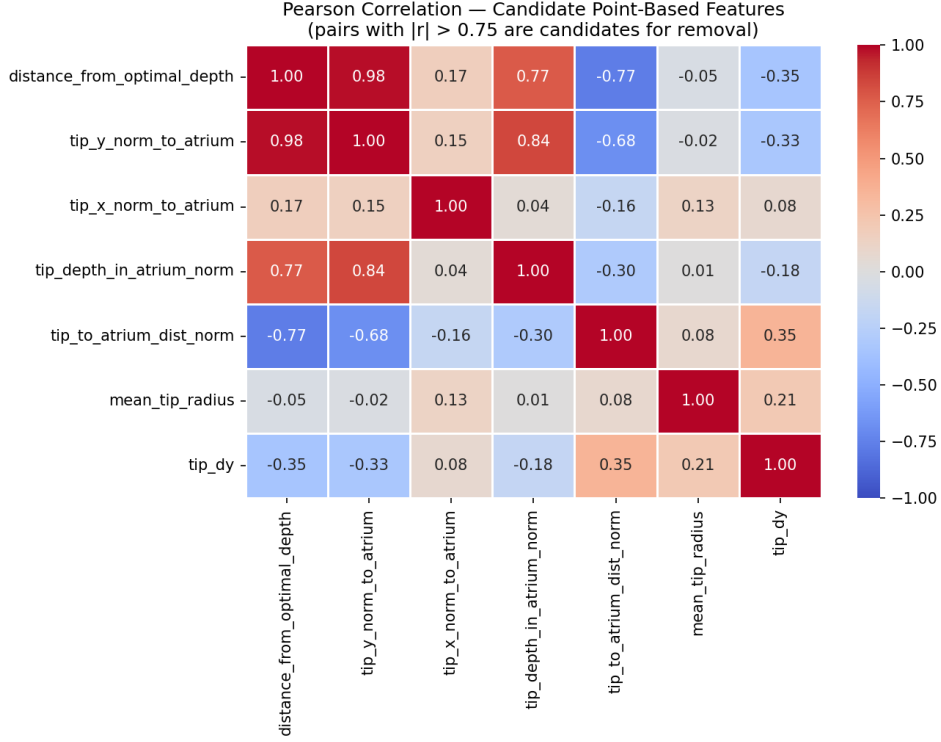


Figure 6: Pearson correlation heatmap for the seven candidate tip-point features. Red = positive correlation; blue = negative. Pairs with $|r| > 0.75$ are highlighted.

The depth-related features in Table 3 (`tip_depth_in_atrium_norm`, `distance_from_optimal_depth`, `tip_y_norm_to_atrium`, and `tip_to_atrium_dist_norm`) are mutually correlated, all encoding the same underlying signal: the vertical depth of the tip within the atrium. Only the strongest and most clinically direct one, `tip_depth_in_atrium_norm` ($r = +0.495$ with the classification label), was retained. `mean_tip_radius` is uncorrelated to all other features ($r < 0.07$). By itself it is no better than random (AUC = 0.46, Table 4), but in combination with depth it increases Random Forest AUC from 0.871 to 0.903 and XGBoost AUC from 0.852 to 0.891 (Table 4; the classifiers are described in Section 4), therefore it was kept as a separate, supplementary signal. The remaining candidates (`tip_x_norm_to_atrium` and `tip_dy`) had correlation close to zero with the classification label and were deleted. This parsimonious two-feature set, with a samples-to-features ratio of 110:1 (220 images, 2 features), keeps the classical models interpretable and limits over-fitting. Figure 7 illustrates the spatial relationship each feature encodes relative to the atrium.

Table 4: Single-feature ablation (5-fold cross-validation). `mean_tip_radius` carries no predictive power on its own (AUC below 0.5) but lifts performance when combined with depth, confirming it contributes a complementary signal rather than redundant information.

Feature set	Model	AUC	F1	Accuracy
Depth only	LR	0.814	0.807	0.827
	RF	0.871	0.784	0.818
	XGB	0.852	0.801	0.832
Radius only	LR	0.558	0.507	0.527
	RF	0.462	0.409	0.455
	XGB	0.462	0.388	0.436
Depth + radius	LR	0.829	0.807	0.823
	RF	0.903	0.783	0.796
	XGB	0.891	0.777	0.800

Ablation uses fixed default hyperparameters ($C = 1.0$, class-weight balanced); the main results (Table 5) use grid-search-tuned settings, so absolute values differ slightly.

Final feature set (samples-to-features ratio 110:1):

Feature	Description
<code>tip_depth_in_atrium_norm</code>	Arc-length tip depth, normalised by atrium height
<code>mean_tip_radius</code>	Catheter thickness at the detected tip (pixels)

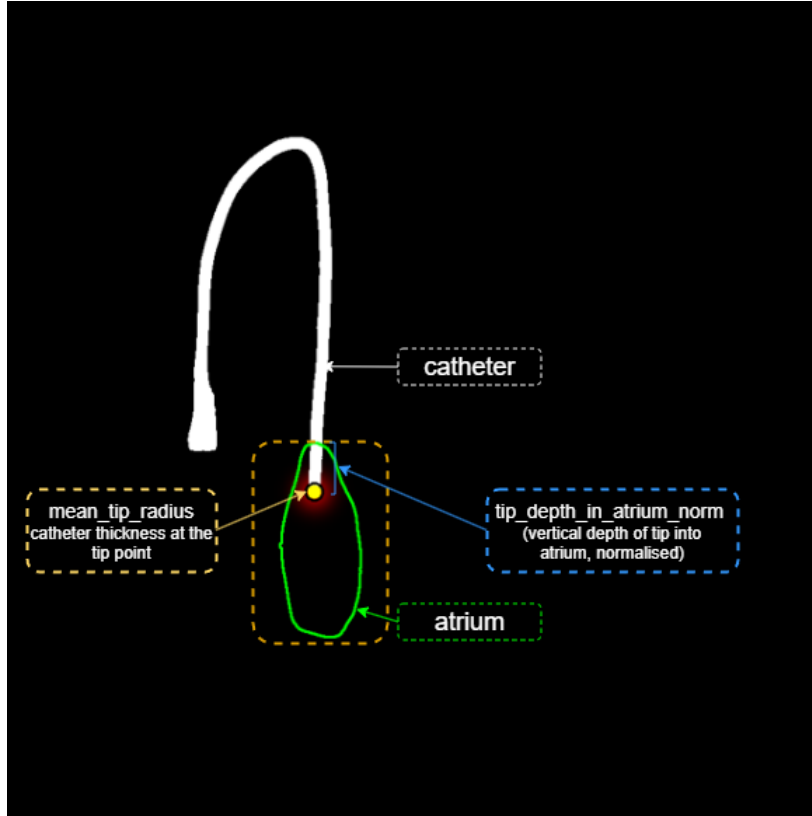


Figure 7: The two selected tip-point features. `mean_tip_radius` is the catheter thickness at the detected tip, and `tip_depth_in_atrium_norm` is the vertical depth of the tip inside the right atrium, normalised by atrium height.

Classical Machine Learning Methods

Experimental Design

Three classifiers were trained using **five-fold stratified cross-validation** (Figure 8) with nested hyperparameter tuning (GridSearchCV [12] on the inner four folds). Stratification ensures that each fold preserves the 54/46 class ratio.

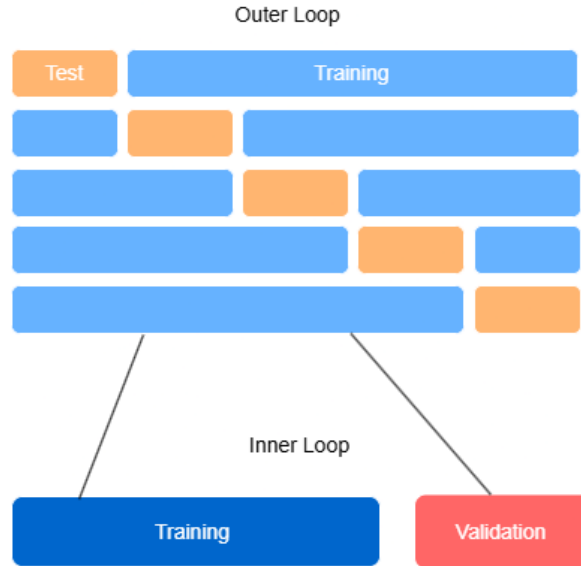


Figure 8: Five-fold stratified cross-validation. The data is split into five folds; in each iteration one fold is held out for validation while the remaining four are used for training, so every sample is used for validation exactly once.

Logistic Regression

Logistic regression fits a linear decision boundary:

$$P(\text{correctly placed} \mid \mathbf{x}) = \sigma(w_1 \cdot \text{tip_depth} + w_2 \cdot \text{radius} + b) \quad (2)$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the logistic sigmoid function. The ℓ_2 -penalised version was used to stabilise coefficient estimates on the small dataset; with only two uncorrelated features, ℓ_2 regularisation shrinks coefficients uniformly toward zero without introducing multicollinearity bias or dropping any feature. The best hyperparameter found by grid search was $C = 0.001$ (strong regularisation), appropriate for a small dataset.

Random Forest

Random Forest [1] is an ensemble of decision trees, each trained on a bootstrap sample of the data. Predictions are made by majority vote across all trees. Best hyperparameters:

max depth = 5, min samples per leaf = 6, min samples per split = 4 (inner CV AUC = 0.895).

XGBoost

XGBoost [3] (extreme gradient boosting) builds trees sequentially, each correcting the residuals of the previous ensemble. Best hyperparameters: max depth = 2, learning rate = 0.03, 200 estimators, $\lambda = 10.0$ (inner CV AUC = 0.890).

Feature Importance

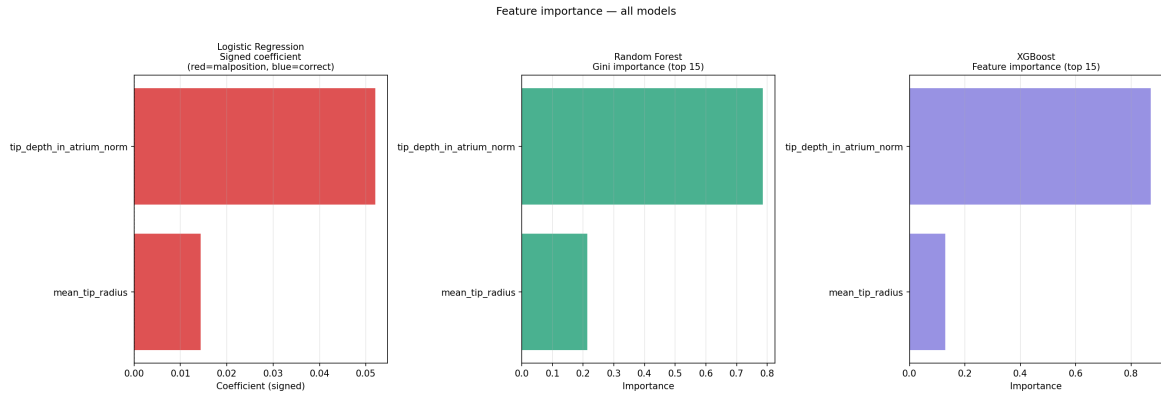


Figure 9: Feature importance across all three classical ML models. Left: absolute standardised LR coefficients. Centre: Gini impurity decrease (RF). Right: XGBoost gain. `tip_depth_in_atrium_norm` is the dominant feature in all three models.

The following metrics are reported throughout. Let TP, TN, FP, FN denote true positives, true negatives, false positives, and false negatives respectively, where *positive* corresponds to correctly placed (label = 1):

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (4)$$

$$\text{F1} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (5)$$

AUC is the area under the receiver operating characteristic curve. All metrics are reported as out-of-fold (OOF) values pooled across all 220 predictions.

Results

Table 5: Out-of-fold performance of classical ML classifiers. SE = std/ $\sqrt{5}$ across folds.

Model	AUC	Accuracy	Sensitivity ($\pm$ SE)	Specificity ($\pm$ SE)	F1
Logistic Regression	0.833	0.814	0.782 ($\pm$ 0.033)	0.840 ($\pm$ 0.025)	0.794
Random Forest	0.900	0.795	0.792 ($\pm$ 0.021)	0.798 ($\pm$ 0.015)	0.780
XGBoost	0.881	0.823	0.772 ($\pm$ 0.042)	0.866 ($\pm$ 0.018)	0.800

Random Forest achieves the highest AUC (0.900) among classical methods, confirming that non-linear interactions between tip depth and tip radius provide additional discriminative power beyond logistic regression.

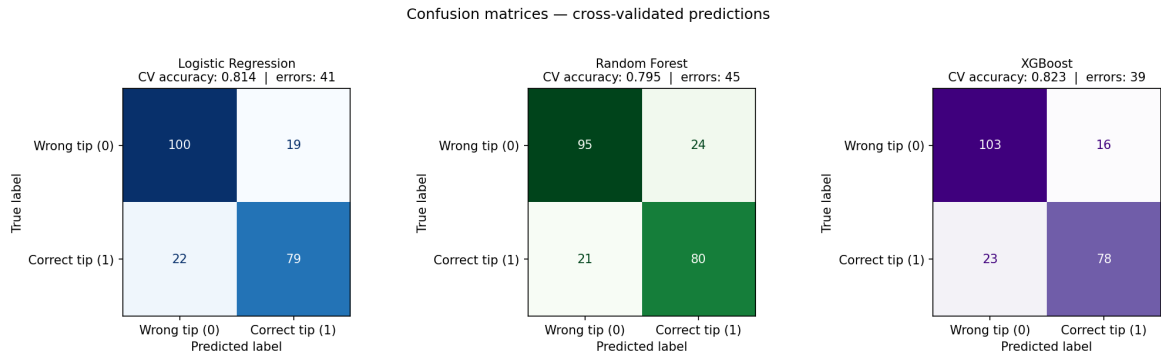


Figure 10: Cross-validated confusion matrices for the three classical ML classifiers.

Table 6 reports the standardised logistic regression coefficients. Both features carry positive coefficients, meaning that a deeper tip inside the atrium and a thicker catheter tip are each associated with a higher predicted probability of correct placement. Both 95 % confidence intervals exclude zero, confirming statistical significance.

Table 6: Logistic Regression standardised signed coefficients ($C = 0.001$, ℓ_2 penalty).

Feature	Coef.	$\pm$ Std	95 % CI	Interpretation
tip_depth_in_atrium_norm	+0.042	0.004	[+0.037, +0.047]	$\uparrow$ correctly placed
mean_tip_radius	+0.012	0.003	[+0.008, +0.016]	$\uparrow$ correctly placed

Since label = 1 corresponds to correctly placed, a positive coefficient means the feature increases the predicted probability of correct placement.

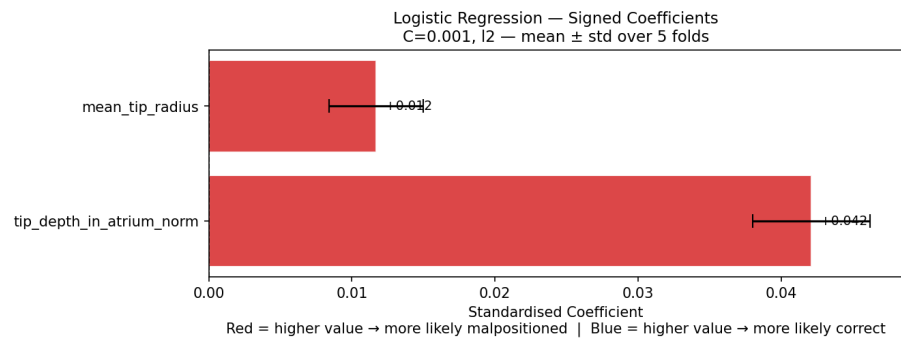


Figure 11: Standardised logistic regression coefficients with 95 % confidence intervals (mean $\pm$ std across 5 folds).

EfficientNet-B7 with Tip-Dot Input

Motivation and Design

The hypothesis underlying the tip-dot model is that the *spatial relationship between the tip location and the right atrium* is the primary discriminating signal for CVC classification. Rather than providing the entire catheter shape to the network, a Gaussian dot is placed at the detected tip coordinate, encoding precisely where the tip is without any irrelevant spatial context from the catheter path.

Each image is represented as a three-channel 600×600 tensor:

Channel	Content	Purpose
0	Gaussian dot at (<code>tip_y</code> , <code>tip_x</code>), $\sigma = 35$ px	tip location
1	Atrium mask (resized to 600×600)	clinical reference
2	Zeros	placeholder

The Gaussian dot is defined as:

$$G(y, x) = \exp\left(-\frac{(y - y_{\text{tip}})^2 + (x - x_{\text{tip}})^2}{2\sigma^2}\right) \times 255 \quad (6)$$

scaled to the target image size. The value $\sigma = 35$ px (at 600×600) was selected to be large enough to be clearly visible after resizing while remaining spatially precise. Figure 12 shows an example input tensor: the Gaussian dot marks the tip location on a plain background, while the atrium mask in channel 1 provides the spatial reference for the network.

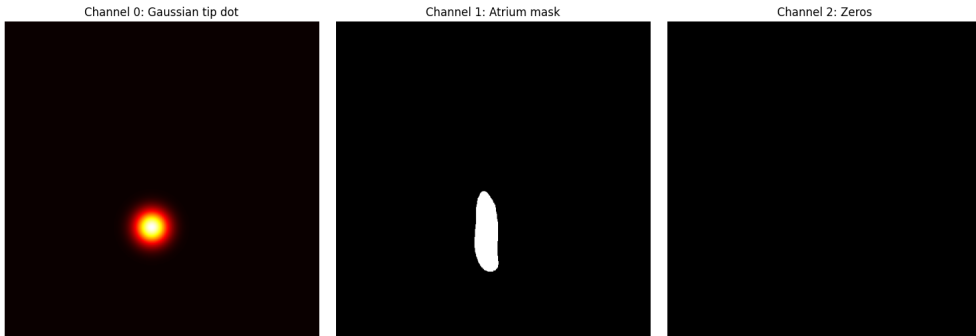


Figure 12: Example three-channel input for the EfficientNet-B7 tip-dot model. Channel 0: Gaussian dot ($\sigma = 35$ px) centred at the detected tip coordinate. Channel 1: resized atrium mask. Channel 2: zeros.

Model Architecture

EfficientNet-B7 is a convolutional neural network with 63.8 million parameters, pre-trained on ImageNet [4] (1.28 million images, 1000 classes). Transfer learning is applied:

- All backbone layers (blocks 0-6) are **frozen**.
- Only the head layers (**conv_head**, **bn2**, classifier) are **trainable**: approximately 1.8 M of 63.8 M parameters (2.8 %).
- The original classifier is replaced by: $\text{Dropout}(p = 0.6) \rightarrow \text{Linear}(2560 \rightarrow 1)$.

Head-only fine-tuning is chosen because the tip-dot input is spatially simple (a blob on a plain background) and does not require the backbone to learn new visual features. With only 220 training images, keeping the backbone frozen also strongly reduces the risk of overfitting.

Training Configuration

- **Loss**: BCEWithLogitsLoss with **pos_weight** = 1.178 (= 119/101, mild class correction).
- **Optimiser**: AdamW [9] ($\eta = 10^{-4}$, weight decay = 10^{-4}) with cosine annealing.
- **Early stopping**: patience 15 epochs on validation AUC; maximum 50 epochs.
- **Input normalisation**: $\mu = (0.5, 0.5, 0.5)$, $\sigma = (0.5, 0.5, 0.5)$ applied to all channels.
- **Data augmentation** [2] (**training only**): horizontal flip ($p = 0.3$), rotation $\pm 15^\circ$ ($p = 0.2$).
- **Cross-validation**: 5-fold stratified, **random_seed** = 42.

Results

Table 7: EfficientNet-B7 tip-dot model per-fold and OOF performance ($n = 220$).

Fold	AUC	Accuracy	Sensitivity	Specificity	F1
1	0.983	0.955	0.950	0.958	0.950
2	0.950	0.886	0.950	0.833	0.884
3	0.863	0.818	0.700	0.917	0.778
4	0.869	0.795	0.750	0.833	0.769
5	0.892	0.818	0.810	0.826	0.810
Mean $\pm$ std	0.911 ± 0.053	0.855 ± 0.066	0.832 ± 0.115	0.874 ± 0.060	0.838 ± 0.077
OOF overall	0.892	0.855	0.832 ($\pm$ SE 0.051)	0.874 ($\pm$ SE 0.027)	0.840

OOF pools all 220 predictions at threshold 0.50. Best epochs: fold 1 = 21, fold 2 = 3, fold 3 = 3, fold 4 = 16, fold 5 = 45. SE = std/ $\sqrt{5}$ across folds.

Table 7 shows that performance varies noticeably across folds. Folds 1 and 2 achieve AUC above 0.95, while folds 3 and 4 are weaker (AUC 0.863 and 0.869). This fold-to-fold variation likely reflects differences in which images end up in the held-out set: folds with a higher proportion of borderline cases - where the tip sits close to the atrium boundary - tend to yield lower AUC. The OOF AUC of 0.892 pools all 220 predictions and is therefore the most reliable overall estimate. Figure 13 shows the resulting confusion matrix at threshold 0.50.

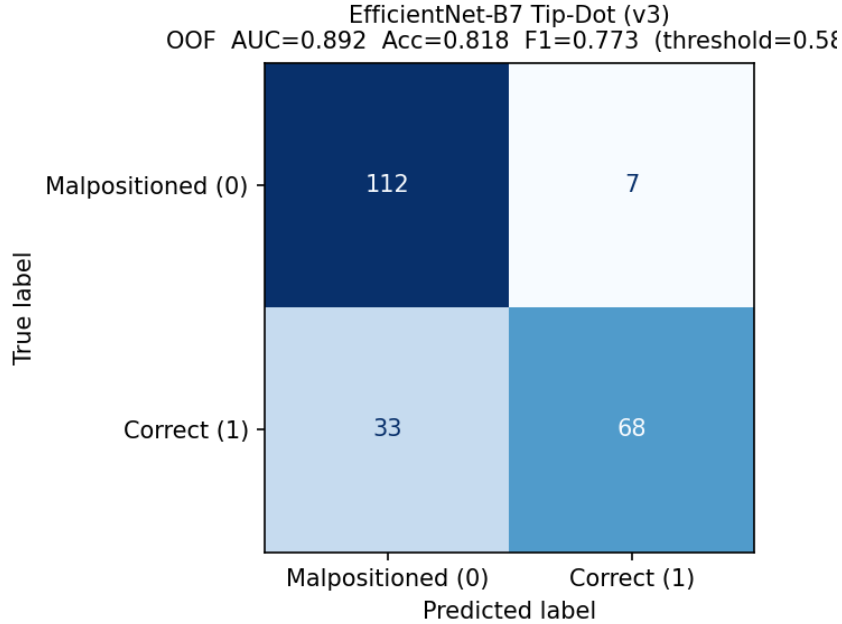


Figure 13: Confusion matrix for the EfficientNet-B7 tip-dot model (OOF predictions, $n = 220$, threshold = 0.58).

Error Analysis by Tip Detection Status

Table 8: EfficientNet-B7 tip-dot performance by tip detection status.

Status	n	Accuracy	AUC	F1
success	197	0.863	0.934	0.830
success_overlap_mode	3	0.667	0.500	0.667
ambiguous_tip	20	0.400	0.791	0.143

*For **ambiguous_tip** cases the Gaussian dot is placed at an unreliable coordinate, degrading performance significantly. These cases motivate the two-step routing strategy.*

EfficientNet-B7 with Full Catheter Mask (Baseline)

Architecture and Input

This model replicates and retraines the EfficientNet-B7 approach of Forgwa (2025) [5].

The three-channel input is:

Channel	Content	Information
0	Full catheter mask (600×600)	complete catheter shape
1	Atrium mask (600×600)	right atrium boundary
2	Zeros	placeholder

Unlike the tip-dot model, this model receives full spatial information about the catheter trajectory, not just the tip location.

Training Configuration

- **Fine-tuning:** last two convolutional blocks plus head - 51.5 M of 63.8 M parameters trainable.
- **Dropout:** $p = 0.3$.
- **Loss:** BCEWithLogitsLoss (no class weighting).
- **Optimiser:** AdamW ($\eta = 10^{-4}$, weight decay = 10^{-3}) with linear warmup (5 epochs) followed by cosine annealing.
- **Epochs:** 60 (no early stopping); best checkpoint per fold selected by validation AUC.
- **No input normalisation** (consistent with Forgwa’s original training).

Results

Table 9: EfficientNet-B7 full catheter mask baseline (Forgwa [5]) - per-fold and OOF performance ($n = 220$). Per-fold sensitivity and specificity were not recorded during training; OOF values are computed from saved prediction arrays.

Fold	AUC	Accuracy	Sensitivity	Specificity
1	0.971	0.886	—	—
2	0.952	0.886	—	—
3	0.942	0.841	—	—
4	0.902	0.795	—	—
5	0.983	0.886	—	—
OOF overall	0.930	0.859	0.890 ($\pm$ SE 0.046)	0.832 ($\pm$ SE 0.037)

No early stopping was used; best checkpoint per fold selected by validation AUC over 60 epochs. $SE = std/\sqrt{5}$.

The full-mask model achieves OOF AUC = 0.930 (Table 9), exceeding the tip-dot model (0.892, Table 7) because it receives complete spatial context about catheter trajectory. However, it does not distinguish between cases where the tip is reliably detected and cases where it is not.

Two-Step Classifier

Design Rationale

The error analysis in Table 8 reveals that the tip-dot model degrades sharply on `ambiguous_tip` images because the Gaussian dot is placed at an unreliable coordinate. The full-mask model, by contrast, does not depend on the tip coordinate at all and is unaffected by tip detection failures.

The proposed two-step classifier routes each image to the model best suited to its tip detection status:

- If `status` $\in$ `{success, success_overlap_mode}` ($n = 200, 91\%$): the tip coordinate is reliable. $\Rightarrow$ Use the **tip-dot EfficientNet-B7**.
- If `status` = `ambiguous_tip` ($n = 20, 9\%$): the tip coordinate is unreliable. $\Rightarrow$ Use the **full-mask EfficientNet-B7**.

The class distribution of the 20 ambiguous-tip cases is reported in Section 7.3 alongside the routing results.

Ensemble Prediction

For both routes (tip-dot and full-mask, as described in Section 7.1), the five fold models trained during cross-validation each produce a probability score for the input image. These five scores are averaged to give a single ensemble probability, which is then thresholded at 0.50 to produce the final binary prediction. Averaging reduces the variance introduced by any single fold’s training outcome.

Two separate input transforms are applied to match each model’s training conditions:

- **Tip-dot route:** normalisation $\mu = (0.5, 0.5, 0.5)$, $\sigma = (0.5, 0.5, 0.5)$.
- **Full-mask route:** no normalisation (consistent with Forgwa’s original training).

Applying the wrong transform to either route would produce corrupted predictions.

Results

Of the 20 ambiguous-tip cases routed to the full-mask model, 13 were correctly placed (label = 1) and 7 were malpositioned (label = 0). The group is genuinely mixed and not dominated by either class, confirming that the $AUC = 1.000$ achieved by the full-mask model on this subset (Table 10) reflects genuine discriminative power rather than a trivial class-imbalance artefact.

Table 10: Two-step classifier performance broken down by tip detection status and the model each status was routed to ($n=220$, threshold = 0.50). Images with reliable tip detection (status `success` or `success_overlap_mode`) were passed to the tip-dot model; images with unreliable tip detection (`ambiguous_tip`) were passed to the full-mask model.

Status	Model	n	Acc.	AUC	Sens.	Spec.
<code>success</code>	Tip-dot	197	0.909	0.963	0.899	0.892
<code>success_overlap_mode</code>	Tip-dot	3	0.667	0.500	0.667	1.000
<code>ambiguous_tip</code>	Full-mask	20	1.000	1.000	1.000	1.000
Overall		220	0.914	0.965	0.931	0.899

Sens. = Sensitivity; Spec. = Specificity. 200 images routed to tip-dot; 20 to full-mask.

Table 11: Two-step classifier - overall OOF metrics ($n = 220$). SE = std/ $\sqrt{5}$ across folds.

Metric	Value
AUC	0.965
Accuracy	0.914
Sensitivity	0.931 ($\pm$ SE 0.033)
Specificity	0.899 ($\pm$ SE 0.015)
F1	0.908

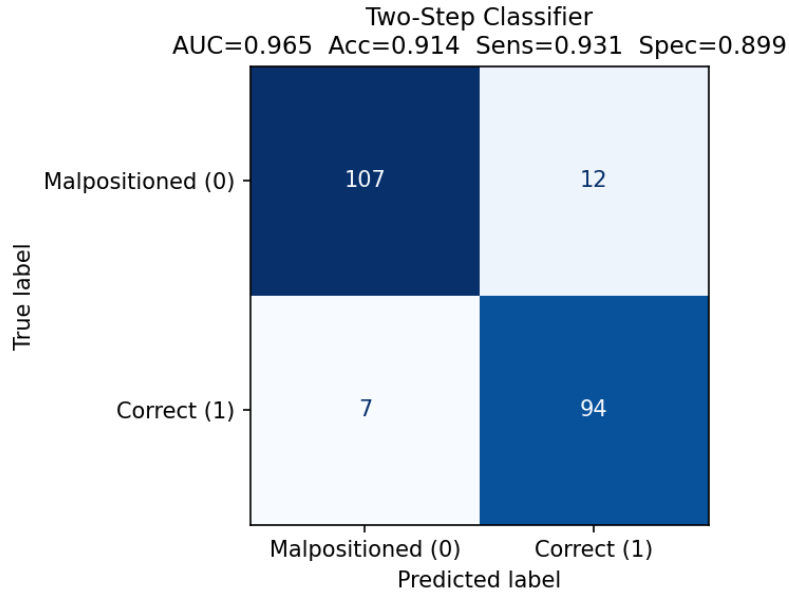


Figure 14: Confusion matrix for the two-step classifier (OOF predictions, $n = 220$, threshold = 0.50).

Full Comparison and Improvement Summary

Table 12: Comparison of all classification approaches. All results are OOF from 5-fold stratified cross-validation ($n = 220$). Best value per column is **bold**. Sensitivity = Recall; Specificity = $TN/(TN+FP)$.

	Method	AUC	Accuracy	Sensitivity	Specificity	F1
ML	LR (2 tip-point features)	0.833	0.814	0.782	0.840	0.794
	RF (2 tip-point features)	0.900	0.795	0.792	0.798	0.780
	XGBoost (2 tip-point features)	0.881	0.823	0.772	0.866	0.800
DL	EfficientNet-B7 tip-dot (proposed)	0.892	0.855	0.832	0.874	0.840
	EfficientNet-B7 full mask (Forgwa, 2025)	0.930	0.859	0.890	0.832	0.853
	Two-step classifier (proposed)	0.965	0.914	0.931	0.899	0.908

SE values are reported separately in Tables 5, 9, and 11.

Table 12 compares all five approaches evaluated in this thesis. Among the classical ML methods, Random Forest achieves the highest AUC (0.900), outperforming both Logistic Regression (0.833) and XGBoost (0.881). Among the deep learning models, the full-mask EfficientNet-B7 (AUC 0.930) outperforms the tip-dot model (AUC 0.892) when evaluated across all 220 images. The two-step classifier achieves the best performance across all metrics, with AUC 0.965 and specificity 0.899. Detailed interpretation of these results is provided in Section 9.

Improvement over the Baseline

	Previous best	Proposed two-step
Method	EfficientNet-B7 full mask (Forgwa, 2025)	Two-step classifier
AUC	0.930	0.965 (+0.035)
Accuracy	0.859	0.914 (+0.055)
Sensitivity	0.890 ($\pm$ SE 0.046)	0.931 ($\pm$ SE 0.033) (+0.041)
Specificity	0.832 ($\pm$ SE 0.037)	0.899 ($\pm$ SE 0.015) (+0.067)
F1	0.853	0.908

The two-step classifier improves AUC by +0.035 and specificity by +0.067 over the full-mask baseline. The improvement in specificity is clinically the most meaningful: the system now correctly identifies 90 of every 100 malpositioned catheters, compared to 83 of every 100 with the previous best model. The smaller standard errors for the

two-step classifier also indicate more consistent performance across data splits.

Discussion

Interpretation of Results

The overall performance gain of the two-step classifier (AUC 0.965, Table 11) over the full-mask baseline (AUC 0.930, Table 9) is due to two complementary effects.

For the 197 images with a reliably detected tip, the tip-dot model (AUC 0.934 on this subset, Table 8) is more discriminative than the full-mask model on the same cases. This is consistent with the hypothesis that the tip location relative to the atrium is the *primary* signal for classification; the complete catheter path adds noise that the model must learn to ignore.

For the 20 ambiguous-tip cases, the full-mask model achieves $\text{AUC} = 1.000$ (Table 10). This result indicates that the complete catheter mask contains rich spatial information that remains discriminative even when the exact tip coordinate cannot be determined.

Even classical ML with only two features (AUC up to 0.900, Table 5) already captures the dominant signal, confirming that tip depth is highly discriminative. The tip-dot EfficientNet (AUC = 0.892 overall) captures additional spatial patterns from the two-dimensional input, including the horizontal position of the dot relative to the atrium edge - information that the single scalar feature `tip_depth_in_atrium_norm` cannot encode. The convolutional feature maps allow the network to encode several of the seven candidate features simultaneously.

This work targets a clinical decision-support setting where specificity - the ability to correctly identify malpositioned catheters - is the primary safety metric. A false negative (system reports “correctly placed” when the catheter is malpositioned) is clinically dangerous: the clinician may begin infusion without repositioning. A false positive (system flags a correctly placed catheter) leads only to an unnecessary re-check. The two-step classifier achieves specificity = 0.899, meaning it correctly identifies approximately 9 out of 10 malpositioned catheters. The reduced standard error (0.015 vs. 0.037 for the baseline) indicates that this performance is more consistent across different subsets of patients.

Placing these results in the context of the broader literature, the two-step approach compares favourably with existing automated CVC assessment systems. Jung et al. [7] applied a similar two-stage pipeline - UNet++ segmentation followed by EfficientNet-B4 classification - to a dataset of 808 radiographs, obtaining a five-fold cross-validated accuracy of 0.76 for a three-class (shallow/proper/deep) problem. Direct numerical comparison with the present work is not straightforward owing to differences in task formulation (three-class vs. binary), input representation (raw radiographs vs. seg-

mentation masks), and dataset size; however, the architectural parallel - segmentation followed by CNN classification - validates the general two-stage design choice adopted here. Hansen et al. [6], evaluating three deep learning models for CVC malpositioning assessment on post-procedure chest X-rays, similarly concluded that incorporating explicit anatomical knowledge into the model design is beneficial, a finding directly aligned with the motivation of the tip-dot approach proposed in this thesis. A consistent theme across the literature is that no single study uses an identical evaluation protocol, making cross-study numerical comparison unreliable; external validation of the proposed pipeline on a larger, multi-centre cohort is therefore an essential next step before clinical deployment.

Limitations

Small single-centre dataset

The 220 images originate from a single hospital (Białystok Medical University). Performance on images from other institutions, different X-ray equipment, or different patient populations is unknown. External validation on an independent multi-centre dataset is required before clinical deployment.

Dependency on segmentation quality

The entire pipeline depends on the quality of the UNet++ segmentation masks. Segmentation errors propagate into tip detection errors, which in turn affect classification. In 9.1 % of cases, the segmentation was ambiguous enough that tip detection failed entirely.

Potential inter-rater variability

The classification labels were assigned by radiologists at Białystok Medical University based on conventional anatomical landmarks. No inter-rater agreement measure is reported in the dataset documentation. Disagreement between annotators on borderline cases would introduce label noise that cannot be quantified or corrected in this study.

Frontal images only

All 220 chest X-rays are posterior-anterior (frontal) views. Lateral radiographs provide complementary depth information about catheter tip position that is not available here. A system trained and evaluated solely on frontal images may miss depth-related malpositions that are only visible in the lateral view.

Limited exploration of alternative approaches

This thesis evaluated three classical ML classifiers and one deep learning architecture (EfficientNet-B7). Other segmentation backbones, alternative deep learning architectures, and end-to-end learned tip localisation methods were not explored. It is possible that a different architecture or a fully learned feature representation would outperform the hand-crafted tip-point approach used here.

No end-to-end training

The tip detection algorithm and the EfficientNet models are trained independently. An end-to-end approach where tip detection error could be back-propagated through the classifier might improve performance.

Future Work

1. **Larger and multi-centre datasets:** Validate on images from multiple hospitals and scanner types to assess generalisability.
2. **End-to-end training:** Jointly optimise the tip localisation and classification objectives.
3. **Uncertainty quantification:** Provide prediction confidence intervals to indicate when the model is uncertain and human review is most needed.
4. **Real-time deployment:** Package the pipeline as a DICOM-integrated tool for bedside use in the radiology or ICU workflow.
5. **Extended label types:** Distinguish between subtypes of malposition (too shallow, too deep, lateral) rather than the binary classification used here.

Ethical Thinking, Societal Relevance, and Stakeholder Awareness

Ethical Aspects

Data and patient privacy

The dataset includes chest X-ray images of dialysis patients from Białystok Medical University (Poland). Images were acquired as part of standard clinical care. The data were submitted to the study group in pseudonymised form: each photograph is identified by an opaque alphanumeric code (`ap_id`) and there is no name, date of birth or other directly identifying information in the analysis files.

Within the General Data Protection Regulation (GDPR), pseudonymised medical images are nevertheless considered personal data because re-identification is theoretically conceivable using imaging biomarkers (e.g. rib geometry). All data were only handled on secure university computing infrastructure (VSC Leuven) and were not shared outside the research group. A GDPR checklist was filled in and submitted according to the UHasselt Master Thesis Guidelines.

Informed consent

The chest X-rays were taken for clinical, not research, purposes. Using clinical images from previous studies for secondary research analyses is the accepted ethical procedure at medical universities. Ethical consent for use of the data was granted by the original data collecting institution (Białystok Medical University) as part of the collaboration with UHasselt.

Honest reporting

All results in this thesis are reported from genuine cross-validated evaluations without cherry-picking. The limitations of the dataset - in particular its single-centre nature and small size - are clearly stated. No results are fabricated or misrepresented.

Algorithmic fairness and bias

The model was trained on images from a single institution in Poland. Chest X-ray appearance varies with scanner manufacturer, exposure settings, and patient body habitus. Performance may differ systematically for patients with different body types, anatomical variants, or images acquired on equipment not represented in the training data. These potential sources of algorithmic bias must be investigated before clinical deployment.

Societal Relevance

CVC misplacement is a preventable patient safety incident. Studies estimate that CVC-related complications affect approximately 15 % of placements globally, with a subset attributable to malposition [10] (McGee & Gould, 2003). In resource-limited settings or during out-of-hours periods, specialist radiologist coverage may be unavailable for hours. An automated classification tool could:

- Reduce the time between catheter insertion and verified safe use.
- Provide a safety net when specialist review is delayed.
- Support less experienced clinicians in interpreting the radiograph.

The societal benefit is directly proportional to the classifier’s *specificity*: a missed malposition harms the patient, so the primary objective of the system is to maximise the detection of unsafe catheter positions while keeping false-alarm rates at an acceptable level.

Stakeholder Awareness

Patients: The primary beneficiaries. Correct and timely identification of malposition reduces complications, hospital stay, and mortality.

Radiologists and clinicians: The direct users. The system is designed as a *decision-support tool*, not a replacement for clinical judgement. Any borderline or flagged case should still be reviewed by a qualified clinician.

Hospital administrators: Reduced complication rates translate to cost savings. However, integration with existing DICOM and PACS infrastructure requires investment.

Regulatory bodies: In Europe, AI-based clinical decision-support tools are regulated as medical devices under the EU MDR (Medical Device Regulation 2017/745). CE marking would require rigorous clinical validation beyond the scope of this thesis.

Researchers: The Forgwa–Bhowmick pipeline provides a reproducible baseline for future CVC classification research on this dataset.

Conclusion

This thesis proposed and evaluated a two-step deep learning pipeline for automated classification of CVC tip position from chest X-ray segmentation masks.

The main contributions are:

1. A skeleton-based **tip identification algorithm** that successfully finds the free end of the catheter using a thickness rule, including loop recovery and loop-vs-artefact distinction. The algorithm found the tip in 200 of 220 images (91 %).
2. **Tip-point feature engineering**: seven candidate features calculated from the detected tip coordinate. By correlation-based selection, two features were retained (**tip_depth_in_atrium_norm** and **mean_tip_radius**). Classical ML models using only these two features reached AUC up to 0.900 (Table 5), confirming the discriminative potential of tip location.
3. A **tip-dot EfficientNet-B7 model** that encodes the tip location as a Gaussian overlay on the atrium mask, achieving OOF AUC = 0.892 overall and 0.934 on reliably detected tips (Table 7).
4. A **two-step classifier** that routes images based on tip detection reliability, combining the complementary strengths of the tip-dot and full-mask models. The system achieves OOF **AUC = 0.965**, **accuracy = 0.914**, **sensitivity = 0.931** and **specificity = 0.899** (Table 11), improving over the full-mask baseline by +0.035 in AUC and +0.067 in specificity.

The most clinically important finding is the gain in specificity: the proposed system correctly identifies 9 out of 10 malpositioned catheters compared to approximately 8 out of 10 for the prior best model, with lower cross-validation variability.

Future work should focus on external validation on a larger, multi-centre dataset and integration into a real-time clinical workflow.

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Generative AI Acknowledgement

I used Claude (Anthropic) for the following purposes during this thesis project: LaTeX formatting and layout improvement, code debugging assistance, and English language improvement in written sections. All scientific content, methodology design, experimental results, and interpretations are my own work.

Software and Code

All code was written in Python 3.13 and executed on the Vlaams Supercomputer Centrum (VSC, Leuven). The main dependencies are listed in Table 13.

Table 13: Main software dependencies.

Package	Purpose
PyTorch 2.12	Deep learning framework
timm 1.0.27	EfficientNet-B7 pretrained model
scikit-learn 1.9	Classical ML, cross-validation
scikit-image 0.26	Skeletonisation, morphology
OpenCV 4.13	Image loading and processing
albumentations 2.0	Data augmentation
NumPy 1.26	Numerical computation
pandas 3.0	Data management

All source code, figures, and output files are available in the public GitHub repository:

<https://github.com/ApuBhowmick/Master-Thesis>

The repository is organised as follows:

Table 14: Repository structure.

Path	Contents
notebooks/	The two main Jupyter notebooks
figures/	All figures used in this report
outputs/	Saved predictions
data/	Feature CSV and label file
README.md	Setup instructions and notebook overview

The two main notebooks are:

- `catheter_tip_efficientnetB7_only_tip_position.ipynb` - tip detection, feature engineering, classical ML, and tip-dot EfficientNet training.
- `EfficientNetB7_Weights.ipynb` - full-mask model training, the two-step classifier, and the sensitivity/specificity analysis.